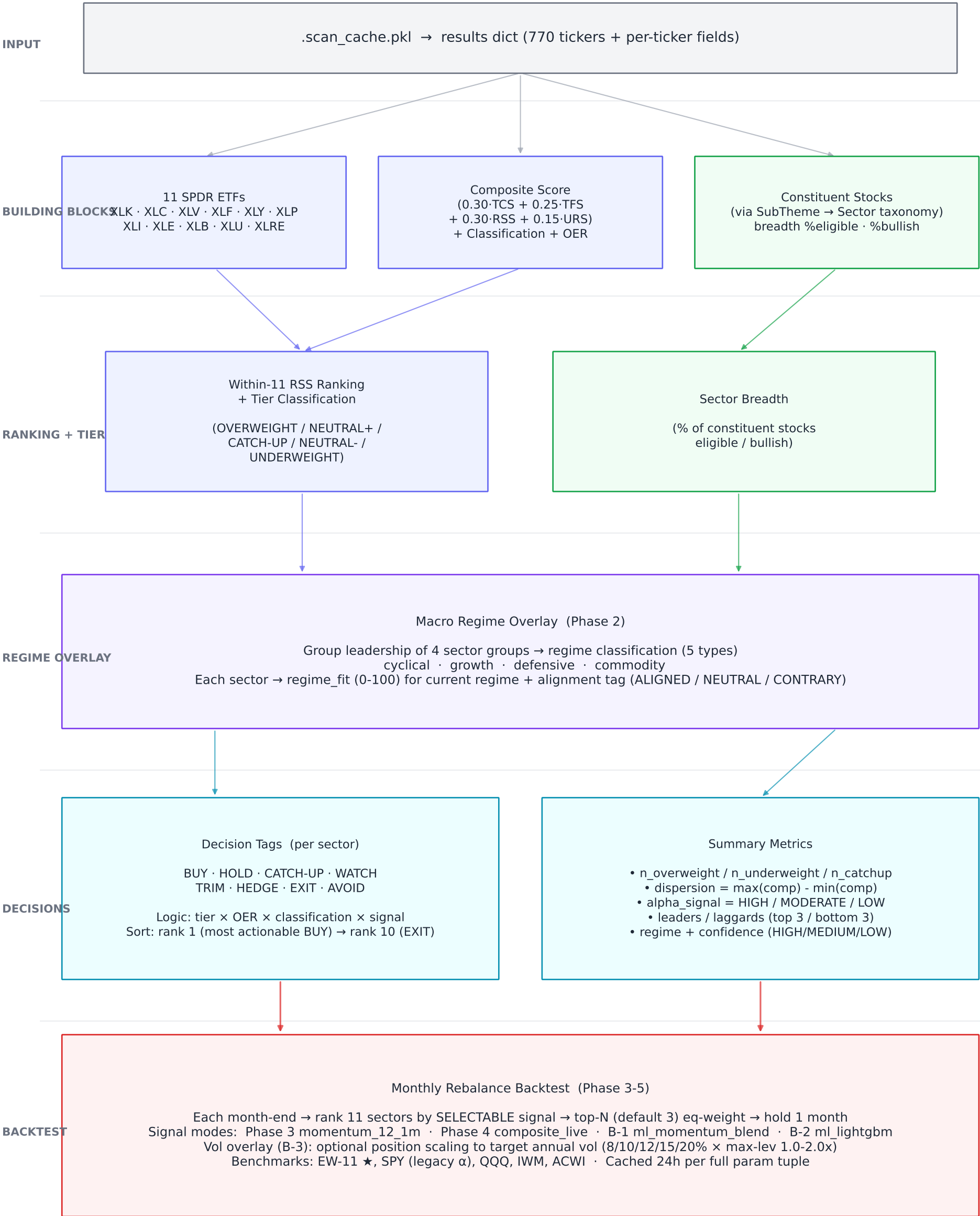


US Sector Rotation — Dependency Graph

11 SPDR ETFs · Tier classification · Macro regime overlay · Monthly backtest



Within-11 ranking + per-sector tier + constituent breadth

Tier Thresholds (5-tier classification)

Tier Classification Logic

OVERWEIGHT : Composite ≥ 70 AND classification ∈ {CONTINUATION, FORMATION, RECOVERY}
(OVEREXTENDED also included if Composite ≥ 70 — but watch OER)
NEUTRAL+ : Composite 55-70 AND bullish classification
CATCH-UP : classification = LAGGING_CATCHUP OR (Composite < 55 AND URS ≥ 70)
→ lagging sector with catch-up potential (high URS = market under-reacting)
NEUTRAL- : Composite 40-55 (borderline / watch)
UNDERWEIGHT : Composite < 40 OR DOWNTREND/CYCLE_PEAK/FADING/WEAKENING
OR (OVEREXTENDED + OER ≥ 75 — severely overextended)

Decision Mapping

Per-Sector Action (long-only — no shorting)

Tier OVERWEIGHT → BUY (rank 2)
Tier NEUTRAL+ → HOLD (rank 5)
Tier CATCH-UP → CATCH-UP (rank 4)
Tier NEUTRAL- → WATCH (rank 6)
Tier UNDERWEIGHT → AVOID (rank 7)

Overrides (precedence):
classification ∈ {DOWNTREND, CYCLE_PEAK}
→ EXIT (rank 10)
OVEREXTENDED + OER ≥ 75 → HEDGE (rank 9)
OVEREXTENDED + OER ≥ 60 → TRIM (rank 8)
FADING / WEAKENING → AVOID (rank 8)

Decision logic is self-contained to the
4-axis scoring model (Composite + class +
OER) — does NOT use hedge-strategy signals.

Sector Breadth (constituent stocks)

For each of the 11 sectors, aggregate
constituent stocks:
pct_eligible : % stocks passing Eligibility Gate
pct_bullish : % in bullish classification
avg_composite : mean composite of constituents
n_constituents : total stocks mapped to sector

Mapping path:
Stock → STOCK_THEMES_CONSOLIDATED → SubTheme
SubTheme → SUBTHEME_TO_SECTOR (api.py) → Sector

Use case: ETF-level signal vs stock-level
breadth divergence — if ETF strong but stock
breadth weak (XLY=55 / 9% breadth), rotation
conviction is lower (only mega-caps driving).

5 regimes × 11 sectors fit matrix · top-down rotation context

Sector Group Classification

4 sector groups (used for regime detection)

cyclical : XLY · XLB · XLI · XLF (aggressive cyclical; early-cycle leaders)
growth : XLK · XLC (Tech / Comm Services; mid-expansion leaders)
defensive : XLP · XLU · XLV · XLRE (Staples / Utilities / Healthcare / RealEstate; risk-off leaders)
commodity : XLE (Energy; late-cycle / inflation hedge)

Regime Detection Algorithm

Detection rules (priority order)

1. Compute average Composite per group (e.g. cyclical_avg = mean(XLY, XLB, XLI, XLF composites))
2. Sort groups by avg descending → leading_group, second_group
3. gap = leading_avg − second_avg

Classification rules:
if cyclical leads AND cyclical > defensive + 8 → RISK_ON_EARLY_CYCLE
elif growth leads AND growth > defensive + 5 → TECH_GROWTH_LED
elif commodity leads AND commodity > 55 AND cyclical > def → LATE_CYCLE
elif defensive leads AND defensive > cyclical + 5 → DEFENSIVE_RISK_OFF
elif gap < 5 → MIXED_TRANSITIONAL
else (fallback): map by leading group's natural regime

Confidence:
gap ≥ 15 AND leading_avg ≥ 55 → HIGH (50 + gap × 2 %)
gap ≥ 8 → MEDIUM (40 + gap × 2 %)
else → LOW (gap × 4 %)

Regime → Per-Sector Fit Matrix (sample)

REGIME_FIT_MATRIX (sector_rotation.py:60+)

Each cell is 0-100 fit score. Higher = sector expected to outperform in that regime.

	XLK	XLC	XLY	XLB	XLI	XLF	XLE	XLP	XLU	XLV	XLRE	
RISK_ON_EARLY		60	55	90	88	88	80	45	30	25	35	40
TECH_GROWTH_LED		95	90	70	45	55	60	35	35	30	55	45
LATE_CYCLE		40	40	35	80	65	60	90	55	60	65	40
DEFENSIVE_RISK_OFF		35	35	20	25	30	30	45	90	90	85	70
MIXED_TRANSITIONAL		55	55	50	50	50	55	50	55	50	60	50

Per-ticker output:
regime_fit : matrix lookup for current regime
composite_x_regime : (composite + regime_fit) / 2 — combined score
regime_alignment : ALIGNED (≥70) · NEUTRAL (50-70) · CONTRARY (<50)

- Practical use:
- ALIGNED + high Composite → strongest conviction (regime + price agree)
 - CONTRARY + high Composite → potential rotation OUT (regime changing)
 - ALIGNED + low Composite → potential rotation IN (early signal)
 - CONTRARY + low Composite → ignore (consistent with regime)

Walk-forward validation · selection alpha vs EW-11 + drawdown reduction vs SPY

Backtest Methodology

Strategy specification

Universe : 11 SPDR sector ETFs (XLK, XLC, XLV, XLF, XLY, XLP, XLI, XLE, XLB, XLU, XLRE)

Benchmarks : EW-11 (★ true selection-α baseline) · SPY · QQQ · IWM · ACWI

EW-11 = equal-weight 11 sectors (isolates SELECTION α)

SPY = cap-weighted S&P 500 (broad US, cap-weight bias)

QQQ = Nasdaq 100 (large-cap growth/tech tilt)

IWM = Russell 2000 (US small-cap)

ACWI = MSCI All Country World (global broad)

Lookback : configurable 3 / 5 / 7 / 10 years

Signal : selectable per request — Phase 3 12-1M momentum / Phase 4 Composite-live

Rebalance : month-end · top-N sectors (default 3) · equal-weight · 1-month hold

Turnover cost : configurable bp (default 30) · applied per position change

Missing data : restrict universe to ETFs with valid forward return AND signal

Data source : yfinance, monthly auto-adjusted Close prices

Computed Metrics

Per-track metrics (× 6 tracks: strategy + 5 benchmarks)

total_return : $\prod(1 + r_t) - 1$

cagr : $(1 + \text{total_return})^{(1/\text{years})} - 1$

sharpe : $(\text{avg} \times 12 - \text{rf}) / (\text{std} \times \sqrt{12})$
(rf = 0 default)

mdd : $\min(\text{cum} / \text{running_max} - 1)$

avg_monthly_ret : mean of monthly returns

vol_monthly : monthly std deviation

Comparison metrics (per benchmark)

alpha_total / annualized : vs SPY (legacy)

alpha_vs_ew_total / annualized ★ : vs EW-11 (selection α)

alpha_vs_qqq_total / annualized : vs QQQ

alpha_vs_iwm_total / annualized : vs IWM

alpha_vs_acwi_total / annualized : vs ACWI

win_rate_pct : % months > SPY (legacy)

win_rate_vs_ew/qqq/iwm/acwi_pct : per-benchmark win rate

turnover_avg_per_rebalance : avg position changes yearly

★ : per-year ret/α/win across all

Cache: 24h TTL per (lookback, top_n, turnover_bp, signal_mode)

Live Result (5y window · top-3 · 30 bp · 12-1M signal)

Six-track comparison

	STRAT	EW-11	SPY	QQQ	IWM	ACWI	
CAGR	~17.1%	~15.3%	~17.0%	~19.1%	~13.3%	~14.8%	
Sharpe	~1.07	~1.03	~1.07	~0.97	~0.69	~0.99	
Max Drawdown	~-12%	~-18%	~-24%	~-33%	~-27%	~-25%	★ best DD
α vs benchmark (CAGR)	—	+1.79%	+0.12%	-1.98%	+3.87%	+2.29%	
win rate	—	52%	41%	45%	55%	46%	

Reading: best vs IWM (+3.87%, small-cap weak), beats EW/ACWI/SPY, underperforms QQQ (-1.98%) — tech mega-cap concentration in QQQ was hard to match.
Strategy excels in DEFENSIVE-ROTATION bear cycles (2022 +28.7% α vs EW); lags narrow tech bulls.

Daily-reconstructed TCS/TFS/RSS/URS · same weights as live engine

Why this exists

Motivation

Phase 3 used 12-1M momentum as a single-axis proxy because historical Composite is not persisted per-month. Phase 4 reconstructs a Composite-equivalent at each historical month-end directly from daily prices, applying the same logic as the live engine (UPDATED 2026-05: TFS residualization + OER penalty added — Live-engine parity):

base

=

0.30·TCS + 0.25·TFS_resid + 0.30·RSS + 0.15·URS

composite

=

base − 0.10·max(0, OER−40)

Both modes are exposed via /api/sector-rotation/backtest?signal_mode=...

Per-axis reconstruction (simplified vs live)

Components and their data-driven simplifications

- TCS (0.30) Trend Continuation
 - SMA 20 / 50 / 200 distance (above = +, below = − ; weights 8 / 8 / 12)
 - SMA 21-day slope (positive = + ; weights 7 / 7 / 8)
- TFS_resid (0.25) Trend Formation, residualized vs TCS [NEW 2026-05]
 - 60d range position (low → high mapped 0 → 30) + 5d-vs-21d acceleration
 - Cross-sectional OLS: TFS_resid = clip(50 + (TFS − (a + b·TCS)), 0, 100)
 - Removes TCS-TFS information overlap (Live-engine parity)
- RSS (0.30) Relative Strength
 - Cross-sectional percentile of 5d / 21d / 63d / 252d returns (.10/.20/.30/.40)
 - Within-sector RSS hybrid: degenerate (11-sector universe IS the sector)
- URS (0.15) Underreaction — held NEUTRAL (50)
 - LeadLag/AttnGap/Drift/Dispersion need richer data than daily Close
- OER (penalty) Overextension Risk [NEW 2026-05]
 - sma20_dist (>8: +15, >5: +8), sma50_dist (>15: +35, >10: +25, >5: +12)
 - RSI(14) (>80: +25, >70: +12), pct_from_high (>−2: +15)
 - Composite penalty: −0.10·max(0, OER−40) (max −6 points)

Empirical comparison (5y window, top-3, 30 bp)

Composite-live performance — post 2026-05 update (TFS_resid + OER penalty)

	Strategy (composite_live)	EW-11 baseline	SPY benchmark
CAGR	~12.4%	~15.5%	~17.6%
Sharpe	~0.81	~1.05	~1.10
Max Drawdown	~-17.7%	~-18.3%	~-24.0%
Total Return (5y)	~100%	~135%	~160%

Insight: OER penalty introduces defensive bias — overheated sectors automatically demoted. Strategy MDD now LOWER than SPY (−17.7% vs −24.0%), reflecting risk-aware sector selection. → Live-engine parity achieved; backtest now accurately mirrors current production signals.

Walk-forward fit · top-3 vs bottom-3 spread maximization · L2 toward 12-1M-favored prior

Why this exists

Motivation

Phases 3 & 4 use FIXED single-axis signals. Different momentum horizons capture different regime dynamics (1M = short reversal, 3M/6M = trend formation, 12-1M = trend continuation). ML signal blends all four and learns the optimal combination from history with walk-forward CV. Goal: regime-adaptive signal that adjusts when one horizon stops working.

Per-month construction

Walk-forward fit at each rebalance date t

- Build 4 cross-sectionally percentile-ranked momentum features at t (per sector, in $[0,1]$):
m1_skip1 = pct_rank($P[t-1] / P[t-2] - 1$) — 1M short reversal signal
m3_skip1 = pct_rank($P[t-1] / P[t-4] - 1$) — 3M momentum
m6_skip1 = pct_rank($P[t-1] / P[t-7] - 1$) — 6M momentum
m12_1 = pct_rank($P[t-1] / P[t-12] - 1$) — Jegadeesh-Titman 12-1M
(= momentum_12_1m baseline)
- Training window: trailing 36 months (min 18)
- Build features at τ + forward return at $\tau+1$ for each historical τ
- Optimize w ($\sum w=1$, $w \in [0,1]$) via SLSQP to MAXIMIZE:
obj(w) = AVG τ [mean(top-3 fwd_ret) − mean(bottom-3 fwd_ret)]
− 0.10 × $\sum (w - \text{prior})^2$ (L2 toward baseline)
- ★ Safeguard: if optimizer's solution train-score < prior train-score → fall back to prior (guarantees ML never UNDERPERFORMS baseline within training)
- Apply final w to current month features → ranked signal → top-3 selection

Prior: [m1=0.0, m3=0.0, m6=0.0, m12_1=1.0] (PURE 12-1M = momentum_12_1m baseline)

Empirical comparison (top-3 · 30 bp · same data)

Phase 3 vs Phase 4 vs Phase 5 (B-1) — across lookback windows (C-1 expansion)

		momentum_12_1m	composite_live	ml_momentum_blend	ML active%
5y (N=71):	CAGR	17.13%	10.42%	17.13% ✓	0/71
	α vs EW-11	+1.79%	−4.92%	+1.79% ✓	
10y (N=131):	CAGR	12.91%	10.42%	12.91% ✓	0/131
	α vs EW-11	+0.99%	−4.92%	+0.99% ✓	
20y (N=251):	CAGR	10.11%	—	10.11% ✓	0/251
	α vs EW-11	−0.17%	—	−0.17% ✓	(incl. GFC)
max (N=329):	CAGR	8.50%	—	8.88% (slightly different selection)	
(1998-2026)	α vs EW-11	−0.24%	—	+0.03%	0/329

→ Across ALL window lengths up to 28y (~330 monthly obs × 11 sectors), ML fits fall back to prior 100% of the time. Sample size IS the constraint — but baseline is also already well-tuned. Long-term realistic CAGR ≈ 8-10% with MDD ≈ −41% (incl. 2008).

Design rationale & next iterations

Findings from C-1 sample expansion (1998-2026, ~28y)

- Sample size 4-5x ↑ (130 → 330 obs), but ML still doesn't beat prior (12-1M)
- Long-term backtest reveals true MDD (−41% in 2008 GFC) vs short-window (−18%)
- Long-term α vs SPY +0.22% (max) vs −1.12% (10y) — DD compounding helps over time

Implications for further ML:

- Pure momentum signal alone is unlikely to beat 12-1M even with more data
- Lift will come from NEW INFORMATION: macro features (B-2), regime conditioning (B-4), or volatility targeting (B-3, doesn't change selection but improves Sharpe)

Maintenance: re-render this PDF after edits to sector_rotation.py / sector_rotation_backtest.py · python3 draw_sector_rotation_graph.py

Scale strategy exposure to maintain target annual vol · improves Sharpe in vol-bursty regimes

Mechanism

Each month-end, scale position size based on trailing realized vol

1. Compute realized_vol from last 6 months of strategy net returns:
realized_vol = std(monthly_returns_last_6m) × √12 × 100 (annualized %)
2. Compute scale = clip(target_vol / realized_vol, 0, max_leverage)
target_vol = user-set target (8/10/12/15/20% annualized)
max_leverage = user-set cap (1.0 = no leverage; up to 2.0)
3. Apply scale to monthly portfolio:
Equity portion: scale × (top-N equal-weight positions)
Cash portion: (1 − scale) earns 0%
net_return = scale × gross_strategy_return − turnover_cost
4. Turnover includes implicit cash transitions (Σ|Δw_position| + |Δcash|)
so vol-target adjustments incur cost like any rebalance.

Empirical comparison (10y · momentum_12_1m · top-3 · 30 bp)

Vol-target sweep on baseline 12-1M signal

	CAGR	Sharpe	MDD	α_EW	avg turnover
no vol-target	12.91%	0.89	−18.8%	+0.99%	0.41
target 8% / max_lev 1.0	8.13%	0.76	−14.4%	−3.79%	0.47
target 10% / max_lev 1.0	8.95%	0.78	−15.7%	−2.97%	0.48
target 12% / max_lev 1.0	9.77%	0.80	−17.0%	−2.15%	0.47
target 15% / max_lev 1.0	10.66%	0.81	−18.8%	−1.26%	0.45
target 12% / max_lev 1.5	10.71%	0.80	−18.6%	−1.21%	
target 15% / max_lev 1.5	11.65%	0.81	−19.8%	−0.27%	

Reading: in this 10y window strategy already had Sharpe 0.89, so reducing equity exposure (low target) hurts CAGR without large Sharpe lift. Vol targeting shines when (i) strategy has high MDD vs Sharpe, (ii) macro vol regimes shift sharply.

When to use vol-targeting

Decision matrix

- USE vol-target IF:
 - baseline Sharpe < 0.7 AND vol regimes shift sharply (e.g., 2008 GFC inclusion)
 - portfolio is core holding requiring smooth equity curve (behavioral persistence)
 - drawdown reduction more important than absolute return
- AVOID vol-target IF:
 - baseline Sharpe > 0.9 (improvement marginal, CAGR cost real)
 - short investment horizon (vol-target cash drag compounds)
 - already using vol-aware signals (composite_live URS captures dispersion)
- Combinable with any signal_mode: orthogonal mechanism (separate from selection).

Walk-forward gradient boosting with momentum + macro features (VIX/yield/credit/DXY)

Why this works (when momentum-only doesn't)

Information addition vs information rearrangement

B-1 reblends 4 momentum horizons — same information, different combination → no lift.
B-2 adds 7 NEW macro features capturing market regime → lift comes from new information, not from better blending of existing signals. Macro features dominate LightGBM importance (60-80% combined gain), with momentum features adding marginal lift.

Features (per sector × month)

12 input features → 1 prediction (next-month sector return)

Momentum (4): m1_skip1, m3_skip1, m6_skip1, m12_1 (cross-sectional pct rank)

Macro (7): vix — CBOE VIX level
 vix_chg21 — 1M change in VIX
 yield_10y — 10-year Treasury yield
 yield_curve — 10y — 13w T-bill spread
 credit_proxy — HYG return — TLT return (1M)
 dxy — US dollar index level
 dxy_chg21 — 1M change in DXY

Categorical (1): sector_id — integer encoding of XLK/XLC/.../XLRE

All macro features lagged 1 month at training (use month t-1 row to predict t→t+1 return).

Walk-forward fit

At each rebalance date t

- Build training panel: tau ∈ [t − 60, t − 1], features at tau + forward return at tau+1
- Min 36 train periods × ~9-11 sectors = 320-660 training observations
- Fit LightGBMRegressor(n_estimators=80, max_depth=4, num_leaves=15, lr=0.05, reg_alpha=0.1, reg_lambda=0.1, min_child_samples=10, feature/bagging_fraction=0.8, bagging_freq=2)
- Predict next-month return for each sector at t → top-N selection
- Fall back to 12-1M momentum if training data insufficient (early periods)

Empirical result (10y window · top-3 · 30 bp)

ml_lightgbm vs other modes

	CAGR	Sharpe	MDD	α_EW	win_EW	
momentum_12_1m		12.91%	0.89	−18.8%	+0.99%	51%
composite_live		10.42%	0.69	−17.7%	−4.92%	42%
ml_momentum_blend (B-1)	12.91%	0.89	−18.8%	+0.99%		51% (= baseline)
ml_lightgbm (B-2)	★ 16.04%	0.96	−24.6%	+4.12%		62%

Avg feature importance (gain-based, % of total):
vix_chg21 16.1% · vix 14.7% · yield_curve 13.5% · credit_proxy 12.9% · dxy_chg21 12.1%
yield_10y 11.3% · dxy 10.4% · m12_1 2.9% · m6/m1/m3 ~1.5% each · sector_id 1.6%

→ +3.13% CAGR, +0.07 Sharpe, +3.13% α vs EW-11 over 10y. Trade-off: MDD worse (−24.6%) due to occasional concentrated regime bets. Pair with B-3 vol target for smoother equity.

Top of Sector Rotation tab: regime banner · master sector table · pairs trade

Layout (top → bottom in tab)

Section ordering (long-only — pairs/short-leg components removed)

① Regime Banner ② Tier Legend ③ Master Sector Table ④ Backtest section (Page 10)

① Regime Banner

Top-of-tab macro regime indicator

Shows current regime label + confidence (HIGH/MEDIUM/LOW) + leading group + group_avg gap.
Color-coded: cyclical/growth = green/cyan, defensive = orange, commodity = yellow, mixed = gray.

② Tier Legend

5-tier classification color codes (per sector_rotation.py)

OVERWEIGHT (green) : Composite ≥ 70 + bullish class — STRONG conviction
NEUTRAL+ (cyan) : Composite 55-70 + bullish — moderate hold
CATCH-UP (purple) : LAGGING_CATCHUP class OR (Composite < 55 + URS ≥ 70)
NEUTRAL- (gray) : Composite 40-55 — borderline / watch
UNDERWEIGHT (red) : Composite < 40 OR DOWNTREND/CYCLE_PEAK/etc.

③ Master Sector Table — 16 columns × 11 rows

Column reference (sortable; SectorRotationTab.tsx)

#	row index 1-11
Rank	overall rank 1-10 (composite-derived)
Ticker	XLK / XLC / XLV / XLF / XLY / XLP / XLI / XLE / XLB / XLU / XLRE
Sector	human-readable sector name (Tech, Financials, Healthcare, ...)
Group	cyclical / growth / defensive / commodity (Phase 2 grouping)
Comp	Composite score 0-100 (0.30·TCS + 0.25·TFS + 0.30·RSS + 0.15·URS)
Fit	regime_fit 0-100 — sector's expected fit for CURRENT regime (Phase 2 matrix)
Comp×Fit	composite_x_regime = (composite + regime_fit) / 2 — combined score
Class	3×3 classification (CONTINUATION / RECOVERY / OVEREXTENDED / etc.)
OER	Overheating / Exhaustion Risk 0-100 (≥60 = warning, ≥75 = severe)
Tier	5-tier label (OVERWEIGHT / NEUTRAL+ / CATCH-UP / NEUTRAL- / UNDERWEIGHT)
Decision	BUY / HOLD / CATCH-UP / WATCH / TRIM / HEDGE / EXIT / AVOID (overrides: DOWNTREND→EXIT, OVEREXTENDED+OER≥75→HEDGE, ≥60→TRIM)
Br% Elig	% of constituent stocks passing Eligibility Gate
Br% Bull	% of constituent stocks in bullish classification
1D / 1W / 1M / 3M	forward returns at horizons (1, 5, 21, 63 trading days)

Note: 8-strategy hedge engine (Wyckoff/Ichimoku/Darvas/...) is COMPUTED for sector ETFs but NOT used by the rotation strategy — see Decision logic in sector_rotation.py.

Controls · KPI cards · cumulative chart · ML/vol viz · detailed metrics · yearly · positions

Layout (top → bottom in Backtest section)

Section ordering

① Controls bar → ② KPI cards (5) → ③ Mode-conditional viz (vol-scale / ML weights / LGBM importance) → ④ Cumulative return chart (6 lines) → ⑤ Detailed Performance table → ⑥ Yearly Breakdown table → ⑦ Position History by Year

① Controls (5 dropdowns + always-visible vol-target)

Configurable backtest parameters

Top-N sectors: 2 / 3 / 4 / 5 (number of sectors held each month)
Lookback: 3 / 5 / 7 / 10 / 15 / 20 / 25 / Max(1998+) (history window)
Signal: momentum_12_1m / composite_live / ml_momentum_blend / ml_lightgbm
Vol target: Off / 8 / 10 / 12 / 15 / 20 % annualized
Max leverage: 1.0x / 1.25x / 1.5x / 2.0x (only visible when vol-target on)
(Turnover cost fixed at 30 bp · monthly rebalance, month-end day)

② KPI Cards (5 alpha-vs-benchmark headlines)

All α values are CAGR difference (annualized)

α vs EW-11 ★ TRUE selection α — vs equal-weight 11-sector basket (no cap-weight bias)
 α vs SPY — vs cap-weighted S&P 500 (broad market, Tech-tilted)
 α vs QQQ — vs Nasdaq 100 (large-cap growth/tech)
 α vs IWM — vs Russell 2000 (US small-cap)
 α vs ACWI — vs MSCI All Country World (global broad)

③ Mode-conditional viz (one of three appears)

Auto-renders based on signal/vol-target combination

Vol-target ON → Realized-vol + position-scale time-series chart (orange line + cyan line)
signal=ml_momentum_blend → Stacked bar chart of weight history (12-1M cyan / 6M amber / 3M pink / 1M gray)
+ train spread % overlay
signal=ml_lightgbm → Average feature importance bar chart (macro orange / momentum cyan)
showing which of 12 features dominate gain across all walk-forward fits

④ Cumulative Return Chart

Line chart: 6 series

Strategy (cyan, thick) · EW-11 (purple, dotted) · SPY (gray, dashed) · QQQ (amber, dashed) · IWM (pink, dashed) · ACWI (green, dashed)
Y-axis: cumulative % · X-axis: monthly date

⑤ Detailed Performance Table & ⑥ Yearly Breakdown

⑤ Detailed Performance

Rows: Total Return / CAGR / Sharpe / MDD / Avg Monthly / Vol Monthly
Cols: Strategy + each visible benchmark
(each cell shows value + Δ vs strategy)

Strategy column highlighted cyan;
 Δ values colored green/red.

⑥ Yearly Breakdown

One row per calendar year:
Year · #Mo · Strategy · EW-11 · SPY · QQQ · IWM · ACWI · α vs EW · α vs SPY · α vs QQQ · α vs IWM · α vs ACWI

Compounded yearly return ($\prod(1+r)-1$).

⑦ Position History by Year (★ replaces old 'last 12 months')

Full backtest history grouped by year — scrollable max-h 600px

Per year: bold cyan summary row (year · #months · total changes · YTD% strat / YTD% SPY / YTD α)
Below: monthly rows Date · Positions (e.g. 'XLK,XLE,XLU') · $\Delta\#$ (n_changes) · Strat % · SPY % · Alpha
Years sorted descending (latest first). Sticky header. Use to inspect specific historical regimes.